CARRY PROTOCOL

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ABSTRACT

Carry introduces an innovative advertising protocol tailored for the gaming industry, offering the potential for increased revenue for game developers and players. By replacing traditional advertising platforms, this protocol not only enhances the advertising effectiveness for advertisers but also reduces associated costs.

The Carry Protocol introduces a transformative approach to integrating advertising into the world of web3 gaming. Through the innovative concept of slots - virtual spaces, soundscapes, or digital entities within games - it offers an unintrusive avenue for presenting advertisements. These slots, inherently tied to specific game environments, can manifest as visual displays, auditory cues, or even interactive virtual entities. Given their intrinsic value, slots are not just static elements but tradable assets within the Carry ecosystem. The paper delves deep into the mechanics of slots, elucidating their lifecycle, potential monetization strategies through auctions, and the importance of performance metrics in optimizing their impact. Moreover, the role of the Carry token and Carry fuel underpins the economic dynamics of the system, ensuring seamless transactions within the Carry marketplace. This protocol harnesses the power of blockchain and decentralized governance, promoting a balanced, sustainable, and player-centric advertising ecosystem in the gaming world.

1 INTRODUCTION

Carry introduces an innovative advertising protocol tailored for the gaming industry, offering the potential for increased revenue for game developers and players. By replacing traditional advertising platforms, this protocol not only enhances the advertising effectiveness for advertisers but also reduces associated costs.

The Carry Protocol innovatively bridges the worlds of gaming and advertising within the web3 space, offering a unique system for showcasing ads within virtual game environments. These advertising spaces, or "slots," are thoughtfully integrated into games, functioning like digital billboards for interactive promotions. This approach aims to create a better and healthier gaming advertising economics.

Moreover, the protocol incorporates a transparent and equitable auction system to allocate advertising space, where the market value of each slot is determined by real-time demand. This system ensures a fair and open process for all participants.

Carry Protocol aims to challenge traditional advertising approaches within the gaming industry by seamlessly integrating advertisements, benefiting game developers, players, and advertisers alike.

1.1 Problems with Game Advertisement Market

- **Failed Incentives:** The dominance of the play-to-earn dual-token economic model in many web3 games, although initially promising, has demonstrated its limitations. Due to its aggressive and singular incentive structure, it often places developers and players in positions of discordant interests, threatening the game's longevity. In the later stages of the vast majority of play-to-earn games, due to the singularity of incentive mechanisms, players can solely profit by selling their held NFT assets. Regardless of the developers' efforts to salvage or enhance the gaming experience, games remain highly susceptible to entering a "death spiral" within a relatively short timeframe.

  For a sustainable future, web3 games necessitate diversified economic mechanisms that foster collaborative wins among developers, players, and other stakeholders.

- **Platform Monopoly:** Developers have grown weary of the "Apple tax", which, as a giant advertisement platform, dominates the web2 advertising system and web2 monetization standards by monopolizing user data assets. The web2 advertising ecosystem primarily comprises four key players: advertisers, media, advertising platforms, and users. Advertising platforms, by monopolizing user data assets, shape the advertising monetization landscape. However, the advancement of privacy technology and blockchain technology has provided a mature technological solution for a fairer advertising protocol.

  Both developers and users have a strong demand for a more equitable and efficient advertising ecosystem. The current dominance of giant advertising platforms has significantly impacted business and innovation efficiency.

- **Inability to Analyze Player Intent:** Despite the transformative potential of web3 in gaming, many developers operate based on speculative insights about player preferences, leading to games that either wane in popularity or serve primarily for asset arbitrage. The heart of this challenge is the absence of effective tools to analyze user behavior within Web3 games. This oversight hinders developers from truly understanding player intent. Notably,

games like Crypto Kitties, which introduced users to the creation and trade of in-game NFTs, underscore the vast possibilities but also the need for more nuanced player behavior insights.

Using the crypto-native game 'Loot' as a case study, it's evident that following its significant traction within the NFT community, over 20 development teams have emerged with a focus on Loot. These teams are diligently working on an array of games rooted in the foundational model of Loot. However, a minimal number of these endeavors have shown tangible advancement. To solve these issues, the developers and players need to have more interaction during its life lifecycle. Native features in web3, for example, predict the market, provide methods for developers to truly interact with players.

As a slogan in the web3 industry goes, "onchain is the new online." Therefore, a multitude of data dimensions can now be easily captured. We need greater insights into users to enhance the efficiency of advertising and gaming.

1.2 Slot Dynamics

At the heart of the Carry Protocol lies the innovative concept of a "slot," a transformative idea designed to bridge the worlds of advertising and on-chain gaming. In its essence, a slot is a unique digital real estate where advertisements can be strategically placed, integrated seamlessly into the virtual environments of games or other digital platforms.

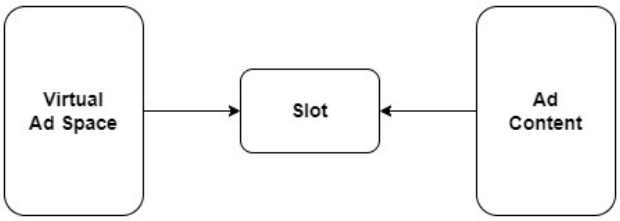

Figure 1: A Slot by Design

- Every slot is inherently tied to a designated game, taking forms such as visual spaces for graphics, character-specific sound effects, or virtual companions roaming the digital environment.
- Beyond their in-game presence, slots hold asset value and can be traded or transferred among participants.
- Slots can be a carrier of various types of media, videos, pictures etc.

Imagine you're playing an online game, and as you explore, you see ads on billboards or even on items like a character's backpack. These are what we call "slots" in the Carry Protocol. Think of a slot like a piece of virtual real estate where ads can live. These slots are valuable to advertisers and can be bought, sold, or traded just like other virtual items in the game.

Slots are versatile. They can show video ads, become part of the game scenery, or even be part of the outfit your character is wearing. They're designed to fit into the game naturally, so they're interesting rather than annoying.

The worth of these ad spaces can go up or down. For example, a slot right where players hang out the most could be worth a lot because more players will see the ad there, while a slot off in a quiet corner might not be as valuable.

The Carry Protocol is all about changing up the game when it comes to ads. These slots are active parts of the game that can change based on how players interact with them and what advertisers want to try. They're a new way to think about ads in games, making sure that players' experiences come first.

2 CARRY: NEXT GEN PROTOCOL FOR WEB3 GAMES ADVERTISEMENTS

2.1 Big Picture

Imagine playing an online game where you can own and trade parts of the game world. That's what's happening with Carry Protocol's 'slots' in new blockchain games. The Carry protocol architecture is shown in Figure 2

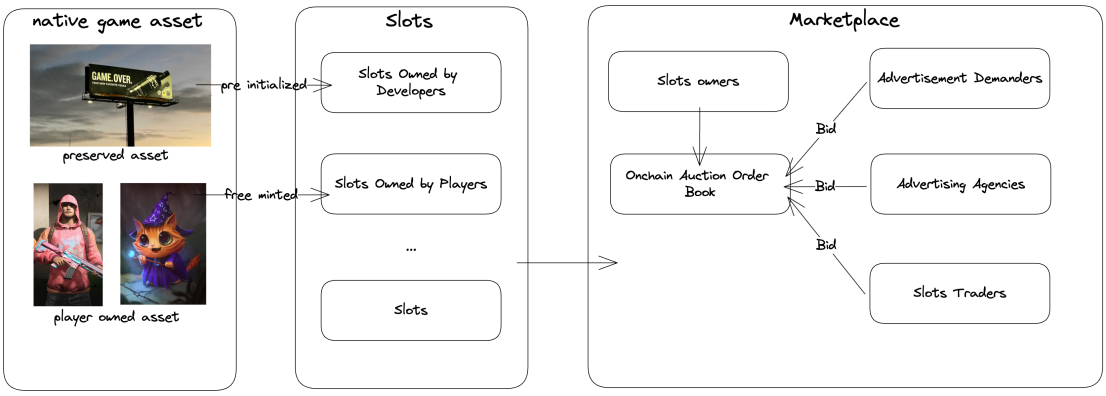

Figure 2: Carry Slot Based Advertising

The Carry Protocol introduces something called "slots" in online games, which are like digital billboards or sponsored items within the game. These slots add value without getting in the way of gameplay and allow advertisers to get more bang for their buck.

Normally in games, the game makers control everything you see and use. But in these new games, players can own unique game items, like a special sword or a piece of land, which they can buy, sell, or trade. These items are called NFTs, which are like digital collectibles that you truly own.

Game developers must establish clear ownership guidelines when introducing slots as in-game advertising spaces. There are two categories of slots: those that are inherent to the game's environment (game native assets) and those that are controlled by players (player owned assets). For instance, a billboard within the game's landscape serves as a game native asset, as shown in Figure 2. Contrastingly, player-specific items such as character skins or pets, which are created and consequently 'minted' by players, represent player owned assets. The economics of these slots, which include their creation, ownership, and value within the game's marketplace, are elaborated in Figure 2.

To elucidate, let's consider a hypothetical scenario. A digital realm, 'LandA', owned by 'Addr1', becomes a canvas for an advertisement. When the developer introduces a slot to this realm, it isn't just assigned to the player 'Addr1'. Instead, it's as if 'LandA' itself becomes enhanced, now comprising both the original realm and the new slot. Symbolically:

$$\text{Addr1} \rightarrow \text{Land A}$$

Transforms to:

$$\text{Addr1} \rightarrow (\text{LandA}, \text{SlotA})$$

This distinction is crucial. The slot isn't an isolated entity but rather correlates with a specific in-game asset. It isn't a generic add-on but a tailored integration. And to facilitate this tailored relationship, technologies like EIP-6220 can be leveraged, ensuring that the connection between the game asset and its corresponding slot is both seamless and efficient.

In summary, Carry Protocol is reimagining how ads fit into video games. Instead of being annoying interruptions, they're becoming a natural part of the game's world. This setup lets game creators and players work together in new ways, making the game more engaging while also figuring out who gets to own these digital ad spaces. It's a fresh approach that could change gaming and advertising for the better.

2.2 Basic Concepts

Within the Carry Protocol, several foundational concepts drive its operation in the web3 gaming world. This section elucidates these essential principles, highlighting their role and importance in the Carry ecosystem.

- **Slot:** The fundamental unit within Carry, serving as the primary canvas for advertisement placements.
- **Slot Time:** A designated time frame for which a slot can be utilized, regulated by the Carry governance.
- **Experimentation:** An integral tool within the protocol facilitating in-depth analysis of on-chain games.
- **Auctioning:** The dynamic process by which slots' time and positioning are bid upon, determining their allocation based on market demand and value perception.
- **Marketplace:** A decentralized platform where slots can be traded, purchased, or leased, fostering a vibrant ecosystem around the advertisement spaces.

2.3 Carry Core

The central tenet of the Carry Protocol hinges on the intricate lifecycle of slots. These slots are not mere placeholders but dynamic entities that embody a unique relationship with gaming assets and advertisements. Here's a closer look at their lifecycle:

- **Creation and Initialization:** Game creators are given special slots that they can easily put into their games. Once these slots are in place, the creators or the players can start using them, turning them into one-of-a-kind digital items. As shown in Figure 3, a billboard in an RPG game is a type of virtual space that can be abstracted as slots.
- **Asset Status Management:** The protocol oversees and modifies the status of each asset, ensuring up-to-date representation within the ecosystem.
- **Asset Relationship Management:** The system smartly manages how different parts of the game work together, particularly when something changes. For example, when a slot stops showing an ad, the ad is no longer connected to that slot.

Figure 3: Example of Advertisement Space in Games: when playing RPG games, the billboard can be abstracted as a slot space where advertisements can be placed on to.

2.3.1 *Slot Status*

- **Idle:** The slot is in a passive state, devoid of any advertisements.
- **Placement:** The slot is actively displaying an advertisement.
- **Pre-initialization:** The slot awaits proper initialization or minting, making it unowned within the ecosystem.
- **Initialization:** Pertains to the formal creation of a slot for a specific game asset, represented technically by the minting of the slot NFT.

2.3.2 *Slot Relationships*

Two primary relationships define a slot's existence:

- **Ownership Dynamics:** Dictates the proprietorship of the slot, tracing it to either a developer, gamer, or directly linking it to a specific in-game asset. At the outset, this ownership typically aligns with a distinct game asset.
- **Advertisement Affiliation:** Emphasizes the bond between the slot and the advertisements it hosts. The intrinsic value of a slot is not just its mere existence, but its capacity to exhibit advertisements and, in turn, generate revenue.

2.3.3 *Slot-Ad Lifecycle*

Examining slots from an advertising temporal perspective helps understand how they accrue value in the ad ecosystem:

- **Slot Placement Duration Auction:** Taking cues from platforms like OpenSea, slots undergo an auction or selection process. Unlike traditional auctions where ownership is

transferred, slots are typically leased for a predetermined duration, conferring advertisement display rights without altering slot ownership.

- **Ad Content Placement:** Once a slot is chosen, it's imbued with the pertinent advertising content.
- **Value Generation:** The advertisement, once viewed or engaged with by users or gamers, starts accruing value.
- **End of Ad Cycle:** Upon reaching the designated period, the advertisement's active phase ceases, reverting the slot to its idle state.
- **Ad Effect Tracking:** This phase is earmarked for gauging the repercussions and reach of the advertisement.
- **Ad Settlement:** Beyond the initial fee amassed during the auction, any supplementary value spawned by the advertisement's efficacy is settled at this juncture.

2.3.4 *Actions on Slots*

Several actions can be taken on a slot, determining its trajectory and interaction with ads:

- **Ad Placement:** Assigning a specific advertisement to the slot for display.
- **Ad Removal:** Extracting the currently showcased advertisement from the slot, returning it to a vacant state.
- **Transfer of Rights:** Conveying the privileges or ownership of the slot to another entity.
- **Slot Exchange:** Engaging in transactions to exchange slots with other participants.

2.3.5 *Placement Strategy*

The strategy of placement ads onto slots determines how slots accrue value and relevance within the gaming ecosystem. As developers and advertisers navigate this space, optimizing placement strategies ensures slots' maximum potential is realized.

Carry provides the following built in strategies:

- **Distinct Time Allocation:** Slots are auctioned for exclusive, well-defined time periods.
- **Periodic Time Allocation:** Slots are systematically scheduled for recurring durations.
- **Enduring Lease:** The slot is granted on a long-term basis to a specific party. Nonetheless, the lessee has the prerogative to auction this lease in the Carry marketplace.

2.4 Carry Governance

Imagine Carry Governance as the manager of a digital billboard. It matches ads with the right spots in the virtual game world, ensuring that everything fits together nicely and the gaming experience stays enjoyable and cohesive. Carry governance:

- **Slot Allocation Governance:** Determines and assigns specific virtual spaces for advertisement displays.
- **Content-Slot Matching:** Regulates and ensures the relevant advertisement content is mapped to its corresponding slot.
- **Ad Lifecycle Management:** Governs the duration and lifecycle of an advertisement within a slot.
- **Ad Quality Oversight:** Establishes standards and ensures displayed advertisements meet quality and relevance criteria.
- **User Feedback Integration:** Incorporates feedback mechanisms and uses player responses to refine ad placements.
- **Dispute Resolution:** Manages and resolves conflicts that may arise in terms of slot assignments or advertisement displays.

2.5 Trading Assets and Using Slots

Carry provides its own marketplace for asset trading/auctioning. These assets include:

- Advertisement Slots
- NFTs
- NFTs that can be used as advertisement content
- Carry token
- Crypto assets like ETH and BTC etc.

Assets can be traded with each other on the Carry marketplace. Slots can be traded through this process. Slot time, based on the strategy, can be auctioned/traded on the Carry marketplace as well.

2.5.1 Auctioning

In the Carry Protocol, slots are like prime digital real estate for ads within games. They are auctioned off to the highest bidder, similar to how you might bid for a popular item on eBay. But instead of buying the slots, advertisers rent them for a certain time. This way, they get to show their ads in the game without owning the slots permanently.

We suggest using a Dutch auction for selling ad space. Here's how it works: the price for advertising starts high and drops over time until someone agrees to pay. Advertisers watch the price fall and jump in when they think the price is right for the value they'll get. This makes the selling process fast and fair because it matches what advertisers are willing to pay with what game developers expect for their ad spaces.

Here's a closer look at the process:

- The slot owner (or their agent) sets an initial price for each slot placement duration. Ideally, this price is higher than the last traded price for the same slot duration.

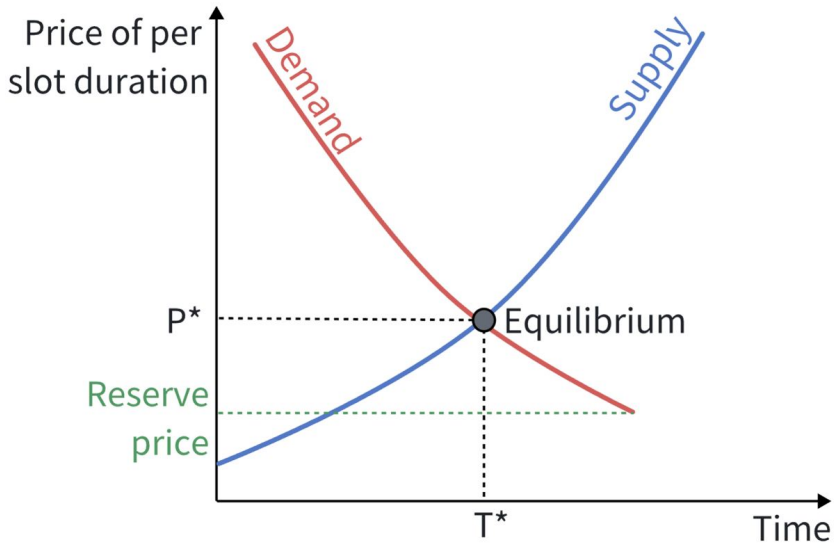

Figure 4: Equilibrium of Slot Pricing

- Given the slot owner's preference for filling their slot durations, the price for the slot placement duration progressively drops over time, urging advertisers to place their bids quickly.
- Advertisers monitor the pricing trend of the slot placement duration. Failing to bid in time and missing out to other bidders means losing an advertising opportunity. Hence, advertisers are inclined to bid once the price seems reasonable.
- Slot owners (or agents) have a reserve price for their slot placement durations. If bids fall below this threshold, the auction becomes void and needs a restart.

2.6 Ad Settlement

Beyond the foundational fee secured during the auction phase, any additional value derived from the ad's performance is settled. The settlement mechanism can be divided into:

1. **Basic Placement Fee (BPF)**:
$$\text{Fee} = \text{BPF} \times t \tag{1}$$
where t is the actual duration of the ad placement.
2. **Performance-based Incentives (PM)**: Depending on the agreement, this could be:
   - **CPS (Cost Per Sale)**: For on-chain businesses, e.g., NFT sales.
   - **CPA (Cost Per Acquisition)**: Based on user metrics.
   - **CPL (Cost Per Liquidity)**: In scenarios like DeFi, based on staked asset value.
   - **CPI (Cost Per Installation)**: Metrics based on app installations.

Ultimately, the revenue (R) for a slot owner is:

$$R = A \times (BPF \times t) + B \times \Sigma PM \times \Sigma PR \tag{2}$$

where PR represents the Performance Ratios, and A and B are coefficients. Slot owners can adjust these coefficients to maximize their returns.

2.7 Slot Lifecycle

In a web3 game, a "slot" is essentially a spot where ads can go, chosen when the game is being made. For instance, in a shooting game, ads could appear as graffiti on a virtual wall. Who owns this ad space depends on where it is. If it's part of the game's environment, like the wall, it's owned by the game's creators. But if it's tied to a player's character, like a logo on a shirt, then the player owns it.

The owner of a slot can decide to use it for ads themselves or offer it up to others by selling or renting it out through a marketplace. This way, slots become a flexible tool for advertising in the gaming world.

3 CARRY ARCHITECTURE & FEATURES

3.1 Carry Slot Contract

3.1.1 Slot Creation

When a new ad campaign is about to start in a web3 game, the first step is setting up a special ad spot, known as a "slot." This slot is made into a unique digital item, like a one-of-a-kind collectible, which means it's owned and can be traded. Game makers weave these slots into their games, and then either they or the players can activate them, turning them into owned digital assets. It's similar to creating a brand-new collectible item that's ready to be paired with an ad.

3.1.2 Slot Release

A slot, once used, may need to be released or made available for future advertisements. This involves the termination or expiration of its current content, making the slot a blank canvas once more. While this could technically be a part of the slot contract, the process's simplicity might not necessitate a full-fledged contract on its own. Instead, it could be an operation under the broader slot management system.

3.1.3 Slot Content Change

Over time, advertisements or the content within a slot might need updating or altering. This could be due to changing marketing strategies, game narratives, or even player preferences. The Carry Slot Contract allows for dynamic content modifications, ensuring that the slot remains relevant and engaging.

3.1.4 Slot Advertisement Placement

The ultimate purpose of a slot is to showcase advertisements, a process termed 'placement' within the Carry Protocol. It's where the rubber meets the road. Once an advertisement is place into a slot, the slot illuminates, fulfilling its purpose. Placement could involve various parameters like the duration of the advertisement, its visual aesthetics, and interactivity levels, all governed and managed by the Carry Slot Contract.

In essence, the Carry Slot Contract is the backbone of the slot system within the Carry Protocol. It ensures that every slot's lifecycle, from creation to placement, is managed efficiently, transparently, and in harmony with the game's environment and the players' expectations.

3.2 Carry Measurement & Metrics

In the dynamic realm of web3 gaming, the Carry Protocol recognizes the value of real-time feedback for in-game advertising. Through Performance Benchmark Metrics, the protocol offers immediate insights into the impact and success of each advertising slot. Carry analysis features can be described by Figure 5.

These metrics don't just offer a retrospective view, they enable active experimentation within games. By analyzing slot performance, developers can refine ad placements, content, and in-

teractivity. This isn't just about placing an ad—it's about optimizing its reach and engagement in a gaming environment.

Additionally, these metrics provide a twofold benefit:

- **For Game Developers:** They get a clear understanding of how integrated advertisements impact gameplay. This ensures that the core gaming experience remains uncompromised while still being monetizable.
- **For Advertisers:** With concrete performance data, advertisers can refine their campaigns. They can tailor content based on which slots garner the most attention, where players spend the most time, and which in-game scenarios drive the most engagement.

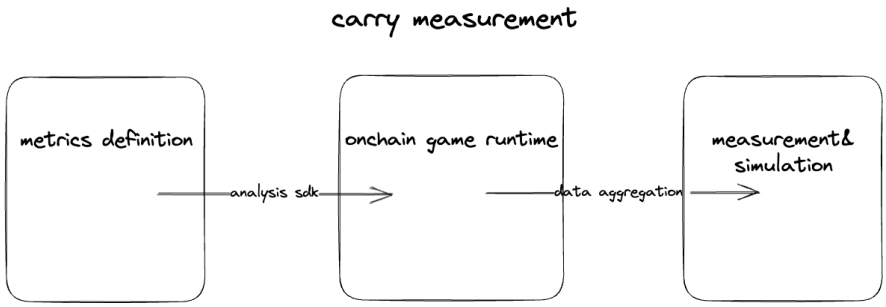

Figure 5: Carry Analysis Structure

3.2.1 *Experimentation*

The Carry Protocol includes a special feature that lets you try out different strategies directly within the blockchain system to see what works best. This tool uses smart contracts, which are like automated rules, to make a secure space for testing. Here, you can change settings and watch how they affect outcomes, helping you make better decisions as you go along.

Core concepts in experimentation for Carry are:

Variables: A variable is the basic unit which strategy makers are trying to measure. Whether it's the slot's visibility, its position in the gaming interface, or the duration of the ad it holds, these variables dictate the behavior of users, regarding certain types of slots. Adjusting them using decentralized logic helps to find the optimal setting for each slot or group of slots.

Groups: Groups can be a collection of any basic unit in the entire carry protocol, cluster of slots, chosen user addresses, gamers in a certain region etc. Groups are a tool for conducting experiments in a scalable way.

Experiments: Each 'experiment' revolves around these slots and their groups. Being smart contracts, they execute tests autonomously on these slot configurations. The experiments can range from testing a new slot position to trying out different ad durations. As these experiments run, they yield data, offering insights into the most effective slot configurations.

Rollouts: Once an experiment concludes, the 'rollouts' come into play. They are the action-driven aftermath of successful experiments, ensuring that proven slot configurations, those variables and group settings that worked, are implemented across the board. It's like upgrading a group of slots based on real, tested data.

In conclusion, experimentation will greatly help users to measure the performance of their strategies. By enabling experimentation, they ensure that the convergence of gaming and advertising is not just seamless but also continually evolving, adaptive, and player-centric.

3.2.2 Metrics

Metrics in the Carry Protocol stand as instrumental indicators, vital in understanding the nuances of in-game advertising in the vast expanse of web3 gaming. Serving as both compass and map, these metrics provide developers and advertisers with real-time insights into the effectiveness and resonance of each ad slot. Notably, these metrics encompass:

- **Actual Views**: Capturing the raw count of users who engage with an ad.
- **Viewer Profiles**: A snapshot of user demographics, interests, and behaviors, allowing for targeted advertising.
- **Conversion Metrics**: Indicators that show the transition of viewers from mere observers to engaged participants or customers.
- **On-chain Payments**: Metrics that display the number of transactions linked to a particular advertisement or slot.
- **Staking Participation**: Data on users who are not just viewing ads but actively staking their tokens or assets.
- **Activation Data**: Metrics highlighting the active engagement and interaction of users with an advertisement.
- **App Installation Data**: Monitoring the number of users who, upon viewing an ad, proceed to install the promoted app.
- **Community Engagement**: Data on users redirected to and actively participating in the advertised community forums or platforms.
- **Social Media Engagement**: Monitoring the traction on social media platforms due to the advertisement, gauging the viral potential of ads.

With these metrics at their fingertips, users can refine their advertisement strategies. They ensure that the ad content doesn't merely inhabit a slot but actively engages, informs, and captivates the gaming audience, optimizing both visibility and impact.

4 ECONOMICS OF CARRY

Within the Carry ecosystem, Carry token plays a role in all aspects of the economics cycle:

Carry Token: The primary currency of the Carry marketplace, the Carry Token facilitates all exchanges that transpire here. From purchasing ad spaces in slots to bidding on prime gaming real estate, the Carry Token is the universal medium. Yet its utility doesn't end merely at marketplace transactions. One of the pivotal functions of the Carry Token is its convertibility: users can exchange it for Carry Fuel, amplifying its significance within the protocol.

Carry Token for Network Burning: Think of Carry Token as the energy driving the actions within the Carry Protocol. Every operation, be it placement advertisements, transferring slots, or toggling slot statuses, consumes Carry Fuel. And once utilized, this fuel is irreversibly burned. The burning mechanism not only ensures transparency and scarcity but also provides a tangible cost to actions, which in turn deters frivolous or malicious behavior on the network.

Carry Governance:The Carry Protocol embraces adaptability through its Carry Improvement Proposal (CIP) mechanism. Carry token holders can initiate and vote on proposals, ensuring that the protocol evolves in line with the community's vision. This democratic approach ensures continuous relevance and responsiveness to industry needs.

The Carry Token establishes an internal economic equilibrium, where actions are not just facilitated but are also assigned real-world value. It's a delicate balance that ensures sustainability, growth, and fairness in the Carry universe.

4.1 Carry Token Utilization Scenarios

Scenatio	Type	Add Value
Slot placement durations auction	Token	Transaction fees discount
Ad Content Placement	GAS	
Slot Initialization	GAS	
Effect tracking	Token	
Slot settlement	Token	
Issue a Proposal	Gvernance	
Vote on snapshot	Governance	Staking Rewards

Table 1: Carry Token Utilizations

5 CARRY STAKEHOLDERS

Carry Protocol's restructuring of the advertising ecosystem redistributes the value that was once dominated by advertising giants back to the advertising ecosystem. This ensures that the industry's genuine contributors receive enhanced incentives, leading to increased efficiency across the entire sector. In this section, we will delve into an analysis of the value that Carry Protocol can deliver to various stakeholders within the gaming industry's advertising ecosystem.

5.1 Developers

Developers play a vital role in the advertising ecosystem. They create services and platforms, providing spaces for ad placements and bridging the gap between users and advertisers.

In many online advertising systems, when advertisers pay to put their ads in games or apps, a big chunk of the money they spend goes to the middlemen—the big companies that control the advertising space. This leaves the creators of the games or apps with less money than they could have earned, which can make them less interested in building new and better things for users.

But with Carry Protocol, the game creators can work directly with the advertisers using something called "slots" for ads, which cuts out the expensive middleman fees. Advertisers pay less, and game creators make more money.

In the old system, game creators didn't always know how well their ads were doing because they had to trust the middleman companies to tell them and to pay them. But Carry Protocol changes that. It lets the game creators see exactly how the ads are performing. With this information, they can be paid fairly based on how well the ads are actually doing.

In the end, Carry Protocol offers a clearer and more honest way of doing things, which should make game creators happy and more willing to get on board.

5.2 Advertisement Demanders

Advertisement demanders are the "financial backbone" of the advertising ecosystem, investing substantial budgets to achieve growth or brand objectives within the industry. A healthy industry should provide valuable feedback to these advertisement demanders.

Despite the significant increase in advertising expenditures due to digital advertising, the most common source of complaints within the industry is advertisement demanders. This is primarily due to the exorbitant fees and non-transparent data mechanisms on advertising platforms, leading to a loss of trust in the industry.

Within the Carry Protocol, we have eliminated the need for advertising platforms by providing slots and an open advertising protocol. This allows advertisement demanders to allocate their budgets directly to services and games offered by developers, thus directly impacting users. This approach significantly enhances the return on investment (ROI) for advertisement demanders and elevates the overall industry efficiency.

To counteract the monopolistic position of advertising platforms, advertisement demanders have traditionally needed to allocate substantial budgets for ad performance evaluations to ensure they are not deceived by the platforms. Due to the dominance of real data by advertising platforms, the process of evaluating ad performance has been quite challenging. In the Carry Protocol, we have developed measurement metrics for assessing ad performance, and both advertisement demanders and developers involved in slot ad placements can access this data transparently. This results in a significant reduction in costs for advertisement demanders when evaluating ad performance and ultimately enhances the efficiency of ad placements.

5.3 Agencies

Advertising agencies play a crucial role in the advertising ecosystem by serving advertisers. Their purpose is to enhance the efficiency of ad placements and increase transparency in the industry. However, in the traditional advertising ecosystem, the data is monopolized by advertising platforms, making it challenging for advertising intermediaries to acquire the necessary data. This, in turn, results in significant difficulties when measuring effectiveness, analyzing audience profiles, and optimizing ad placement strategies.

Advertising agencies are dissatisfied with the current state of the advertising industry. Carry Protocol, as an open advertising ecosystem, eliminates data monopolization by advertising platforms, allowing advertising intermediaries to provide various services to advertisers with the latter's authorization.

Agency's role in the digital advertising ecosystem is important as they act as intermediaries between advertisers (advertisement demands) and slot owners. In a slot duration marketplace, slots are auctioned, and agencies need to devise strategic bidding strategies to secure optimal ad placements for their clients.

5.3.1 Purchasing a Series of Slot Durations in the Marketplace

1. **Slot Selection**: Identify and select a series of slots that align with the advertiser's target audience and campaign goals.
2. **Bid Formulation**: For each selected slot, formulate a bid based on the strategy mentioned above.
3. **Auction Participation**: Participate in auctions for the selected slots, placing bids in real-time.
4. **Winning and Allocation**: If a bid is successful, allocate the slot to the advertiser's campaign. Adjust the remaining budget and strategy for subsequent auctions accordingly.
5. **Performance Monitoring**: Continuously monitor the performance of acquired slots and adjust future bids and strategies based on real-time data.

5.3.2 Dutch Auction (Open Descending Price Auction)

In a Dutch Auction, the price of the ad slot starts high and decreases over time until an advertiser places a bid or the price drops to a reserve price.

AGENCY STRATEGY FOR DUTCH AUCTION: The agency aims to secure the ad slot for its client at the lowest possible price but also needs to balance the risk of losing the slot to another bidder. The strategy could be defined as follows:

$$B_i(t) = \begin{cases} 0 & \text{if } P(t) > V_i \\ V_i & \text{if } P(t) \leqslant V_i \text{ and } \text{Risk}(t, V_i) < \theta \\ 0 & \text{otherwise} \end{cases}$$

Here:

- $B_i(t)$ is the bid from agency i at time t
- $P(t)$ is the current price at time t
- V_i is the agency's valuation for the ad slot
- $\text{Risk}(t, V_i)$ is a function evaluating the risk of losing the slot if waiting any longer, considering the current time and valuation
- θ is a risk threshold, beyond which the agency decides to bid to avoid losing the slot

In a more sophisticated approach, the agency not only considers the current price and its own valuation but also takes into account the time decay, estimated competition, and a dynamic risk assessment.

$$B_i(t) = \begin{cases} 0 & \text{if } P(t) > V_i \\ V_i \cdot \left(1 - e^{-\lambda \cdot (T-t)}\right) & \text{if } P(t) \leqslant V_i \cdot \left(1 - e^{-\lambda \cdot (T-t)}\right) \text{ and } \text{Competition}(t, P(t)) \leqslant \gamma \\ 0 & \text{otherwise} \end{cases}$$

Here:

- T is the total time duration of the auction
- λ is a parameter controlling the time sensitivity of the bid
- $\text{Competition}(t, P(t))$ is a function estimating the level of competition based on the current time and price
- γ is a competition threshold, beyond which the agency decides not to bid due to high competition

In this strategy, the agency increases its willingness to bid as the auction progresses, adjusting its bid according to the time decay function. The agency also evaluates the level of competition at the current price and decides to bid only if the estimated competition is below a certain threshold.

ROI OF AGENCIES' STRATEGY In a Dutch Auction, the price of the advertising slot decreases over time, and agencies aim to secure slots at an optimal price point that balances cost and potential returns. An advanced bidding strategy should therefore consider both the cost of the slot and the expected return on investment (ROI).

The bid function $B_i(t)$ for agency i at time t can be formulated as:

$$B_i(t) = \begin{cases} 0 & \text{if } P(t) > V_i \text{ or } \mathrm{ROI}_i(t, P(t)) < \mathrm{ROI}_{\text{threshold}} \\ P(t) & \text{if } P(t) \leqslant V_i \text{ and } \mathrm{ROI}_i(t, P(t)) \geqslant \mathrm{ROI}_{\text{threshold}} \text{ and } \mathrm{Competition}(t, P(t)) \leqslant \gamma \\ 0 & \text{otherwise} \end{cases}$$

Here:

- $\mathrm{ROI}_i(t, P(t))$ is the expected return on investment for agency i at time t given the current price
- $\mathrm{ROI}_{\text{threshold}}$ is the minimum acceptable ROI for the agency
- $\mathrm{Competition}(t, P(t))$ is a function estimating the level of competition based on the current time and price
- γ is a competition threshold, beyond which the agency decides not to bid due to high competition

The function $\mathrm{ROI}_i(t, P(t))$ can be further defined based on the agency's estimation of the ad slot's effectiveness, the target audience's size, and other relevant factors. On the basis, ROI can be understood as:

$$\mathrm{ROI} = \frac{\mathrm{Income}}{\mathrm{Spending}}$$

In real life the estimation of the real ROI might be more complicated than this. If an agency is conducting one advertisement campaign in one month time, the return can be of the following notation:

$$\mathrm{ROI} = \frac{\Sigma_i(I_i \cdot N_i - E_i) - \Sigma C_{slot}}{\Sigma C_{slot}}$$

where:

- I_i is the item price for i that has been sold in this campaign duration.
- E_i is the expected sales amount of product i.
- C_{slot} is the cost of a slot.

The theoretical returns are a benchmark for the success of any agencies that is on Carry protocol. Real strategy returns can be monitored and calculated through several methods. The Carry measurement and metrics can be used for capturing the real ROIs of any efforts put on the network. Furthermore, we can adopt the P value in the statistics of the sales side to estimate the general impact of certain advertisement campaign.

There are metrics that can also be taken into consideration which agencies could measure when trying to estimate the expected outcome for certain collection of slots:

- **Total Crypto Assets of the Slot Owner's Address:** This reflects the financial strength of the slot owner. Agencies can use this information to assess whether the slot owner has enough resources and motivation to maintain and promote their slots, indirectly affecting the advertisement's effectiveness.
- **Fair Market Price of the Attached NFT to the Slot** This indicates the value of the NFT attached to the slot. High-value NFTs may attract more user attention, potentially increasing the advertisement's exposure and effectiveness. Agencies can use this information to select slots with high-value NFTs for ad placements.
- **Active User Count of the Onchain Game Associated with the Slot:** This reflects the game's activity level and user base size. A higher number of active users means more opportunities for ad exposure. Agencies can prioritize games with a large active user base for their ad placements.

In addition to the above metrics, agencies can choose from a wider range of dimensions available through the Open Carry Ada Network, such as user geographic locations, interests, consumption habits, and more. This allows agencies to offer more precise and personalized ad placement services to advertisers, improving the advertisement's conversion rate and ROI.

By considering a comprehensive set of relevant metrics, agencies can more accurately estimate advertisement effectiveness, providing more valuable and competitive services to advertisers.

5.4 Users

In the traditional advertising ecosystem, users are often the most overlooked participants. Advertising platforms typically assume that since users are already enjoying some basic internet services for free, they should be willing to provide their data and privacy, along with tolerating advertisements. In reality, the commercial value generated from user data has long contributed to the prosperity of the advertising ecosystem. Users, as they engage in internet activities, are not just consumers of services; they actively contribute their data, thereby supporting the ecosystem.

The Carry Protocol believes that users have the right to access the commercial value generated by their data and the ability to choose how their data is utilized. Within the slot framework, the contributions of players and users' data can be transparently tracked, allowing developers, users, and advertisers to negotiate ad settlement terms based on the value of user data.

We encourage developers and advertisers to share a larger portion of the revenue with players and users. This not only increases user engagement but also helps advertisers build more direct relationships with users and foster more adhesive communities, ultimately granting advertisers greater influence over users.

5.5 Comparison of Incentive Distribution

Role	Traditional Ad Platform	Carry Protocol
Developers	Income is limited by platform fees, hindering innovation.	Direct collaboration with advertisers through slots, increased income, reduced costs.
	Lack of transparent ad performance data, unfavorable negotiations.	Access to ad performance data at any time, direct settlement with advertisers.
Advertisers	High platform fees, opaque data mechanisms lower ROI.	Directly allocate budgets to developer services, improving ROI and industry efficiency.
	High cost of ad performance testing, data constrained by platforms.	Use Carry Protocol's measurement metrics for ad performance testing, reducing cost.
Users	Data and privacy are viewed and contributions with limited benefits.	Have the right to receive the commercial value of their data, choose whether data is used.
	Participate in internet activities with contributions but no benefits.	Share more revenue, build more direct relationships and sticky communities.
Agencies	Difficulty in data acquisition limits effectiveness measurement, audience profiling, and campaign optimization.	Provide more services in Carry Protocol's open advertising ecosystem, improving efficiency.

Table 2: Comparison of Incentives

6 CONCLUSION

The Carry Protocol represents a pivotal leap forward in the confluence of gaming and advertising within the burgeoning web3 domain. By introducing a structured and transparent framework for integrating ads into the gaming experience, Carry not only unlocks a new revenue stream for developers but also paves the way for advertisers to reach a dynamic and engaged audience.

Central to this is the concept of slots — dynamic spaces within games that can host various types of advertisements. The Carry Protocol's meticulous design ensures that these slots integrate seamlessly with the gaming interface, ensuring that advertising is both organic and engaging.

The robust governance structure proposed ensures the protocol's continuous evolution, adapting to the ever-evolving landscape of gaming and digital advertising. Moreover, the emphasis on data-driven insights, facilitated through detailed metrics and experimentation capabilities, ensures that every stakeholder, from developers and advertisers to gamers, has the tools they need to extract maximum value from the system.

In a broader spectrum of gaming environments, particularly within fully on-chain games, slots predominantly belong to the gaming community or individual players. Consequently, advertising revenue is more intricately allocated to players and developers. There is a compelling rationale to anticipate that this direction represents the future of advertising.

In essence, the Carry Protocol is not just a technological innovation but a holistic ecosystem, fostering collaboration, innovation, and sustainable growth in the web3 gaming advertisement realm. As the digital realm continues to expand and the lines between gaming and real-world experiences blur further, the Carry Protocol is poised to be at the forefront, shaping the future of in-game advertising.